Solving Linear Systems by Elimination

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Learning Objectives

- Solve systems of linear equations using the elimination method
- Multiply equations by appropriate constants to create opposite coefficients
- Verify solutions by substituting back into the original system

Solve each system of linear equations using the elimination method and write the solution as an ordered pair.

1. Solve the system using elimination.

$$\begin{cases} x + y = 7 \\ x - y = 3 \end{cases}$$

Answer: ______________________

2. Solve the system using elimination.

$$\begin{cases} 2x + 3y = 12 \\ 2x - y = 4 \end{cases}$$

Answer: ______________________

3. Solve the system using elimination.

$$\begin{cases} 3x + 2y = 13 \\ 5x - 2y = 11 \end{cases}$$

Answer: ______________________

4. Solve the system using elimination.

$$\begin{cases} 4x + y = 14 \\ 2x - 3y = 0 \end{cases}$$

Answer: ______________________

5. Solve the system using elimination.

$$\begin{cases} 3x + 4y = 10 \\ 5x - 2y = 8 \end{cases}$$

Answer: ______________________

6. Solve the system using elimination.

$$\begin{cases} 2x + 5y = 11 \\ 3x - 2y = -12 \end{cases}$$

Answer: ______________________

7. Solve the system using elimination.

$$\begin{cases} 4x - 3y = 1 \\ 5x + 2y = 30 \end{cases}$$

Answer: ______________________________________

8. Solve the system using elimination.

$$\begin{cases} 6x + 5y = 7 \\ 2x - 3y = -21 \end{cases}$$

Answer: ______________________________________

9. Solve the system using elimination.

$$\begin{cases} 7x + 2y = 20 \\ 3x - 4y = -6 \end{cases}$$

Answer: ______________________________________

Encourage students to choose the variable that is easiest to eliminate and to always check their answers by substitution.

Solutions

1. Solve the system using elimination.

$$\begin{cases} x + y = 7 \\ x - y = 3 \end{cases}$$

→ *Apply the elimination method to the system [x + y = 7] and [x - y = 3].*
→ *Reduce to one variable and solve to get x = 5.*
→ *Substitute back to get y = 2.*
→ *The solution is (5, 2); it checks in both original equations.*

Answer: $(5,\ 2)$

2. Solve the system using elimination.

$$\begin{cases} 2x + 3y = 12 \\ 2x - y = 4 \end{cases}$$

→ *Apply the elimination method to the system [2x + 3y = 12] and [2x - y = 4].*
→ *Reduce to one variable and solve to get x = 3.*
→ *Substitute back to get y = 2.*
→ *The solution is (3, 2); it checks in both original equations.*

Answer: $(3,\ 2)$

3. Solve the system using elimination.

$$\begin{cases} 3x + 2y = 13 \\ 5x - 2y = 11 \end{cases}$$

→ *Apply the elimination method to the system [3x + 2y = 13] and [5x - 2y = 11].*
→ *Reduce to one variable and solve to get x = 3.*
→ *Substitute back to get y = 2.*
→ *The solution is (3, 2); it checks in both original equations.*

Answer: $(3,\ 2)$

4. Solve the system using elimination.

$$\begin{cases} 4x + y = 14 \\ 2x - 3y = 0 \end{cases}$$

→ *Apply the elimination method to the system [4x + y = 14] and [2x - 3y = 0].*
→ *Reduce to one variable and solve to get x = 3.*
→ *Substitute back to get y = 2.*
→ *The solution is (3, 2); it checks in both original equations.*

Answer: $(3,\ 2)$

5. Solve the system using elimination.

$$\begin{cases} 3x + 4y = 10 \\ 5x - 2y = 8 \end{cases}$$

→ *Apply the elimination method to the system [3x + 4y = 10] and [5x - 2y = 8].*
→ *Reduce to one variable and solve to get x = 2.*
→ *Substitute back to get y = 1.*
→ *The solution is (2, 1); it checks in both original equations.*

Answer: $(2,\ 1)$

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6. Solve the system using elimination.

$$\begin{cases} 2x + 5y = 11 \\ 3x - 2y = -12 \end{cases}$$

→ *Apply the elimination method to the system [2x + 5y = 11] and [3x - 2y = -12].*
→ *Reduce to one variable and solve to get x = -2.*
→ *Substitute back to get y = 3.*
→ *The solution is (-2, 3); it checks in both original equations.*

Answer: $(-2,\ 3)$

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7. Solve the system using elimination.

$$\begin{cases} 4x - 3y = 1 \\ 5x + 2y = 30 \end{cases}$$

→ *Apply the elimination method to the system [4x - 3y = 1] and [5x + 2y = 30].*
→ *Reduce to one variable and solve to get x = 4.*
→ *Substitute back to get y = 5.*
→ *The solution is (4, 5); it checks in both original equations.*

Answer: $(4,\ 5)$

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8. Solve the system using elimination.

$$\begin{cases} 6x + 5y = 7 \\ 2x - 3y = -21 \end{cases}$$

→ *Apply the elimination method to the system [6x + 5y = 7] and [2x - 3y = -21].*
→ *Reduce to one variable and solve to get x = -3.*
→ *Substitute back to get y = 5.*
→ *The solution is (-3, 5); it checks in both original equations.*

Answer: $(-3,\ 5)$

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9. Solve the system using elimination.

$$\begin{cases} 7x + 2y = 20 \\ 3x - 4y = -6 \end{cases}$$

→ *Apply the elimination method to the system [7x + 2y = 20] and [3x - 4y = -6].*
→ *Reduce to one variable and solve to get x = 2.*
→ *Substitute back to get y = 3.*
→ *The solution is (2, 3); it checks in both original equations.*

Answer: $(2,\ 3)$